Editor Guide

GAM200 / CSD2401BF25

Fall AY2025

Owala

Team Roster:

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StageBound
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Summary

This guide breaks down the main components of the Owala Editor interface to help users navigate and utilize its features for game development.

As shown below in Figure 1, the editor has many interface panels that will be explained in this guide.

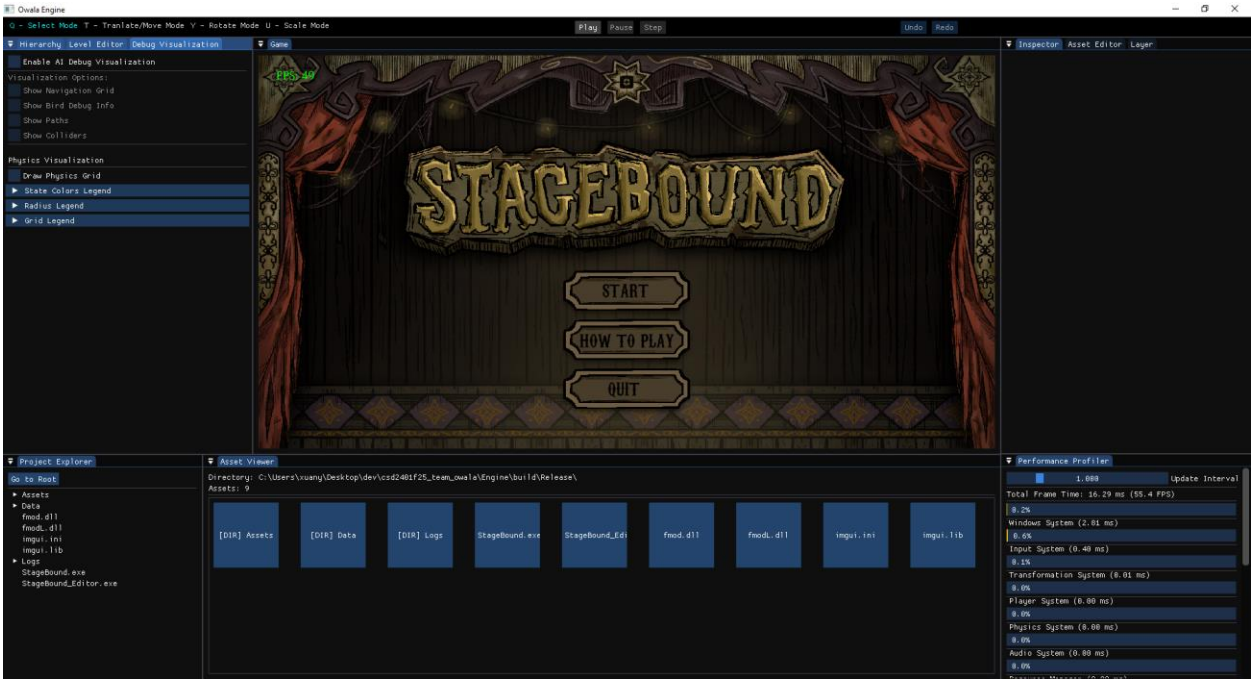


Figure 1: Owala Editor Overview

This guide will explain the functions of each panel and what information they contain, from left to right:

- **Top Left:** Hierarchy, Level Editor, and Debug Visualization panels.
- **Bottom Left and Center:** Project Explorer, Asset Viewer, Tilemap Editor and Prefab Editor panels.
- **Center:** Game Viewport and Toolbar.
- **Top Right:** Inspector, Asset Editor, Layer panels.
- **Bottom Right:** Performance Profiler panel.

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1.0 Scene and Entity Management (Top Left)

This section is crucial for managing the entities within the selected scene.

1.1 Hierarchy

The hierarchy panel displays all entities present within the current scene as shown below in Figure 2.

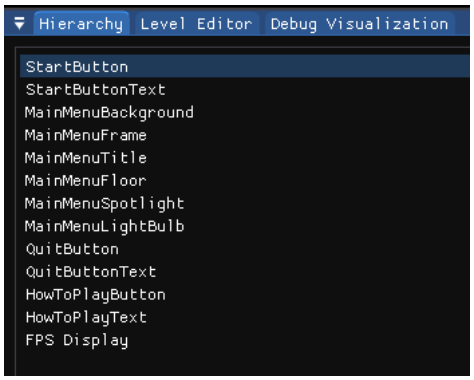


Figure 2: Hierarchy Panel

Each entity in the list can be selected by clicking on it, and it will be shown within the **Inspector** panel on the right side of the editor. There, the entity name, layer, components and tags may be viewed or edited. The selected entity will also be shown under the **World Selection** section in the Level Editor panel.

Right clicking on each entity will also bring out a context menu, where the entity can be copy and pasted, duplicated, or deleted. This can be seen below in Figure 3.

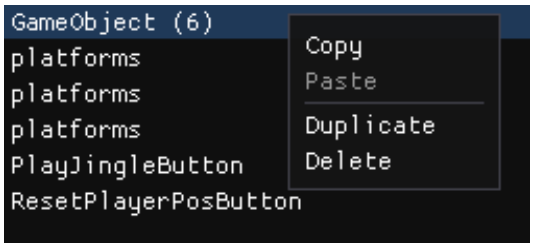


Figure 3: Entity Context Menu

1.2 Level Editor

The Level Editor panel is as shown below in Figure 4.

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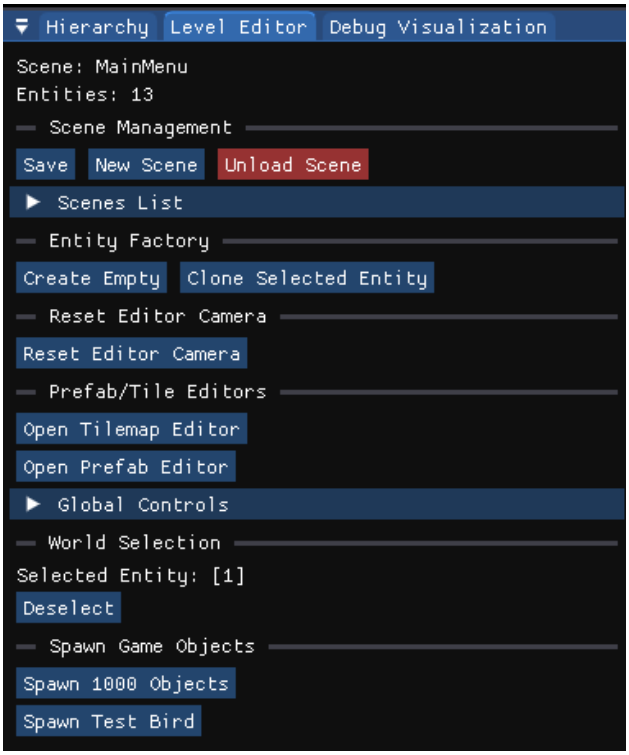


Figure 4: Level Editor Panel

The level editor consists of various functions:

- **Scene Management**
    - Displays current scene name and number of entities present in the current scene.
    - Save button: Save current scene data to JSON file.
    - New Scene button: Create a new scene. This will bring up a modal for user to enter the scene name, which must be unique.
    - Unload Scene button: Unload the current scene. This will bring up a modal for user to confirm if they would like to save the current scene before unloading.
    - Scenes List dropdown: List of scenes currently available to load. Clicking on each scene will load that scene.
- **Open Prefab Editor:** Opens Prefab Editor on the right side of the editor, allowing user to edit the prefabs and save or reload its data from its JSON file.
- **Entity Creation**

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    - Create Empty: Create an empty entity.
    - Clone Selected Entity: Clone the currently selected entity.
- **Reset Editor Camera:** Resets editor camera.
- **Open Tilemap Editor:** Opens the Tile Palette and Prefab Palette panels on the right side of the editor, from which the user can drag and drop objects into the scene.
- **Global Controls**
    - Master Volume: Slider for controlling the global sound level.
    - Show Grid: Toggles visual aids for alignment and selection behavior.
- **World Selection**
    - Selected Entity: Shows entity ID of the currently selected entity. Deselect button allows user to deselect the entity.
- **Spawn Objects/Enemies**
    - Create 1000 Objects: Spawns 1000 entities for stress testing.
    - Spawn Test Bird: Spawn Bird enemy.

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2.0 Debug and Visualization Tools (Top Left)



Figure 5: Debug Visualization Panel

These tools are essential for debugging, performance tuning, and understanding real-time AI and physics behavior. The panel is split into two main sections.

- **AI Debug Visualization:** A dedicated section for visualizing and tweaking Artificial Intelligence components.
    - **Enable AI Debug Visualization:** Turns the feature on/off.
    - **Visualization Options:**
        - **Show Navigation Grid:** Displays the grid used by AI for movement planning. Walkable cells are green and blocked cells are red.
        - **Show Debug Info / Show Paths:** Reveals real-time data about AI entities and their calculated movement routes.
        - **Show Colliders:** Renders the **collision boundaries** of objects, critical for physics and interaction debugging.

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    - **State Colors Legend / Radius Legend / Grid Legend:** Explanations for the various colors and markings used in the debug view.

- **Physics Visualization:** This section provides debug visuals for the core physics engine itself.
    - Draw Physics Grid: Toggles the visibility of the physics engine's broad-phase spatial grid.
        - How it works: This is different from the AI Navigation Grid. This grid is used purely for optimization. Active cells (grid squares that contain at least one collidable object) are highlighted in green.
        - Purpose: This visualizes how the engine optimizes collision detection. By only checking for collisions between objects in the same or adjacent active cells, the system avoids millions of unnecessary checks against objects in empty space.

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3.0 Project Explorer and Asset Viewer (Bottom Left/Center)

These panels allow the user to browse project files within the editor. They are as shown below in Figure 6:



Figure 6: Project Explorer and Asset Viewer

The **Project Explorer** offers a hierarchical view of the project's folders and game assets, similar to a typical file directory tree.

Go to Root: Quickly navigate to the top-level project directory.

The **Asset Viewer** also allows the user to browse the project files and game assets, by clicking each folder button to navigate into the folder and the [DIR] .. button to navigate back up one level. Here, users can view image previews of the game assets.

Within the **Asset Viewer**, users may also click on a file to select it. Upon selection, it will be reflected in the **Asset Editor** panel on the top right side of the editor, where the file can be reimported or replaced. A larger image preview is also available here.

4.0 Tilemap Editors (Bottom Center)

When the Tilemap Editor is opened by pressing the “Open Tilemap Editor” button in the Level Editor, two panels will appear on the top right of the editor: the Prefab Palette and the Tile Palette. On both panels, the Eraser button can be found at the top. Clicking on this button and then clicking on any recently placed prefab or tile will delete that entity from the scene.

4.1 Prefab Palette

The Prefab Palette allows users to level edit efficiently by using prefabs.

Users may left click on a prefab to drag and drop a prefab game object directly into the game viewport, or right click on a prefab to enter Brush Mode.

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In Brush Mode, the prefab will be spawned at the position where the user left clicks within the game viewport.

To erase prefab objects, users can click on the Eraser button and proceed to click on the prefab objects to delete them.



Figure 7: Prefab Palette

4.2 Tile Palette

Similarly to the Prefab Palette, the Tile Palette allows users to build their levels efficiently within the editor by allowing them to place tiles onto the game viewport directly.

The Tile Palette works the same as the Prefab Palette. Users can left click to drag and drop a tile directly into the game viewport, or they can right click on a tile to enter Brush Mode, where they can then place the tiles wherever they left click in the game viewport.

There is also an Eraser button to allow users to delete any tiles that they placed.



Figure 8: Tile Palette

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4.3 Prefab Editor

The Prefab Editor allows users to customize their prefab game objects.

Here, they can edit the prefab properties such as the Mass, Collider size, and Texture Size. This may differ depending on the prefab being edited.

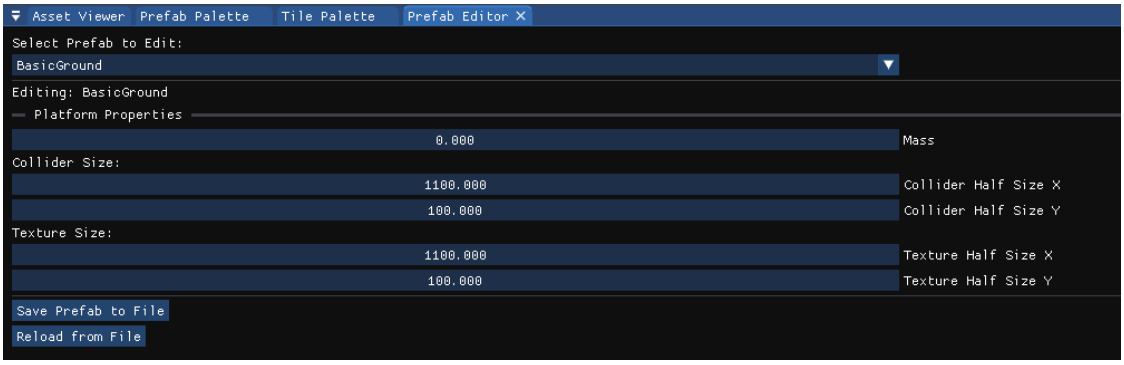
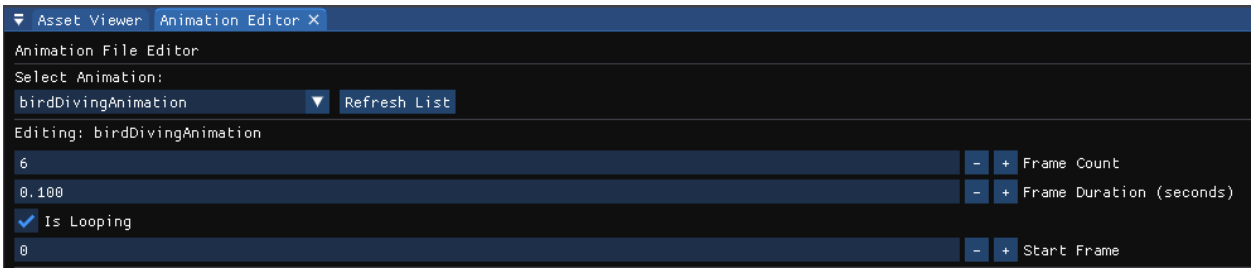


Figure 9: Prefab Editor

After users are done editing the prefab, they can click on the “Save Prefab to File” button to save their changes or they can click on “Reload from File” if they would like to discard their changes and load the original properties back.

4.4 Animation Editor

The animation editor allows users to edit animation related data such as animation frames, animation speed and mapped sprite sheets. After editing, users may press “Save Animation” to save the new properties to disk.



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5.0 Game Viewport and Toolbar (Center)

The Toolbar and Game Viewport in the center of the Editor handle the gameplay simulation. These can be seen below in Figure 10.



Figure 10: Game Viewport and Toolbar

Within the Toolbar above the **Game Viewport**, there are:

- Gizmos Keyboard Shortcuts Information
    - The Gizmos keyboard shortcuts are displayed at the top left corner of the toolbar, with the current mode highlighted in blue to create a better user experience.
- Play/Stop Button
    - This button starts and stops the gameplay simulation. While the simulation is running, the game can be played within the game viewport, allowing the user to test out game mechanics and level design.
    - While gameplay simulation is active, the Play button also turns into a Stop Button. When the user presses Stop button, it will stop the gameplay simulation and reset all entities to their initial state.
- Pause/Step Buttons
    - The Pause button stops the gameplay simulation without resetting all the entities to their initial state. During the paused state, the user can then press the Step button to progress the physics in the game step by step, making debugging easy with this visual aid.
- Undo/Redo Buttons

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- These buttons allow the user to undo an action they have done within the editor, as well as redo it. This feature tracks the past 20 changes made and will be cleared once the scene changes.

6.0 Inspector (Top Right)

The Inspector panel offers a more in-depth view of the currently selected entity and its associated components and tags.

As seen from Figure 11 below, this panel is split into 3 sections: the top section displays the basic entity properties, the middle section displays the components, and the bottom section displays the tags.

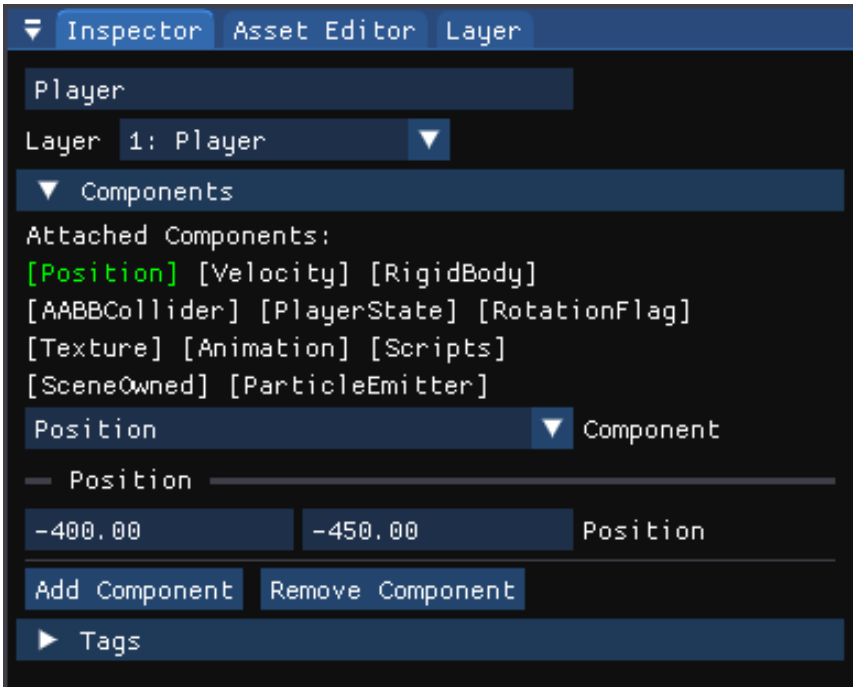


Figure 11: Inspector Panel

6.1 Basic Entity Properties

The top section of the Inspector panel shows the basic properties of the entity, the entity's name and its collision layer. The entity name can be edited, and the collision can be adjusted using the dropdown menu.

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6.2 Components

The middle section shows the components tagged to this entity. It first displays all attached components in a list, and then a dropdown allowing the user to select a component to add, remove, or edit.

As seen from Figure 12 below, selecting the Position component allows for editing the position X and Y values in real time. Clicking on Remove Component will delete this component from the entity.

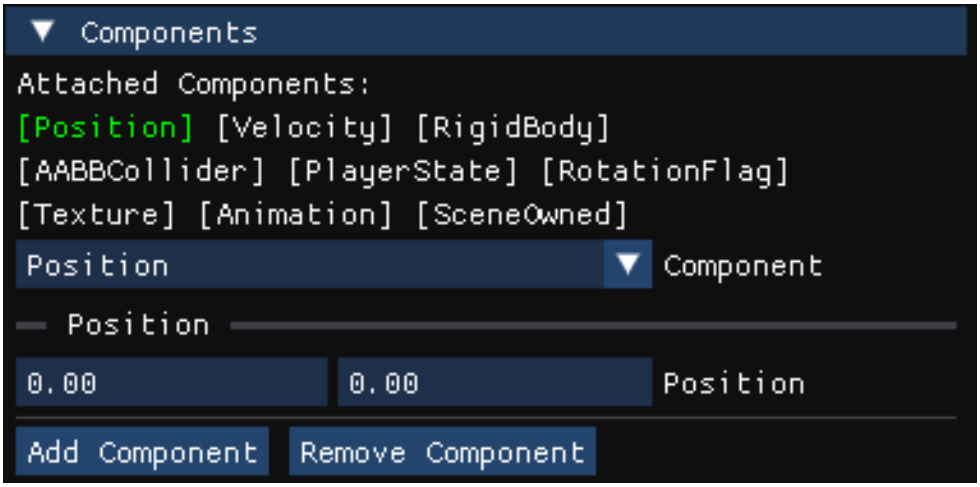


Figure 12: Selecting an attached component

As seen from Figure 13 below, users can freely add any component to the entity.

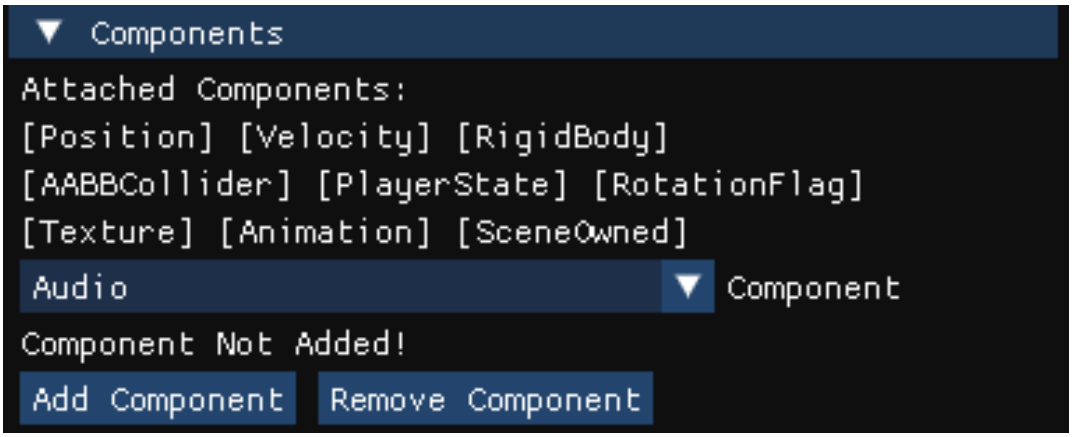


Figure 13: Selecting an unattached component

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6.2.1 Attaching Scripts to a Game Object

Here, adding the Scripts component will allow the user to attach a script to the entity.

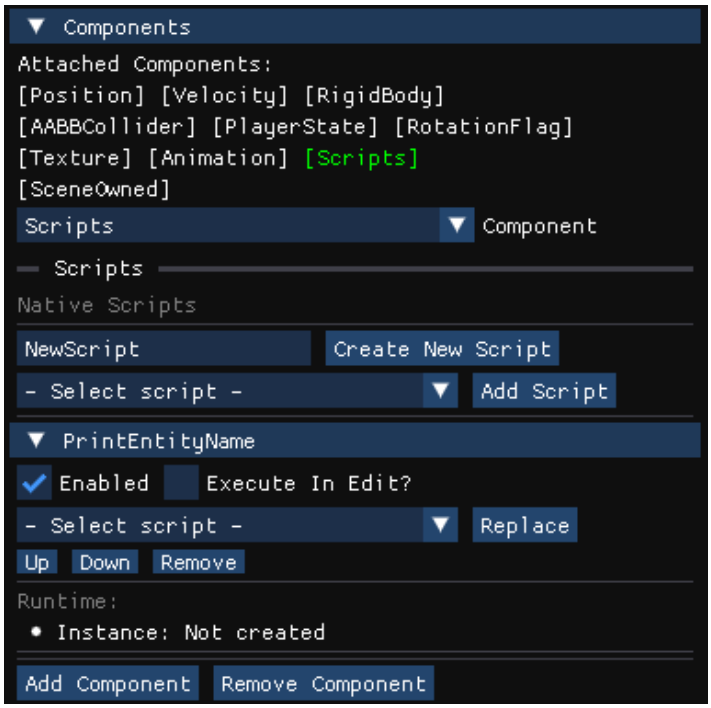


Figure 14: Attaching Scripts

With the Scripts component selected, the user can attach compiled native scripts to the entity. Once attached, the script will run their OnCreate function. Every frame, it will also run its OnUpdate function. In the future, the OnDestroy function will be implemented to run when the component is removed, or when the entity is destroyed.

The user can also choose to add a new custom script by specifying the script name beside the "Create New Script" button, then press the button to create the script. The script (both header and source file) will then be added to the Native Folder under Engine/owala/Native.

Currently, the editor does not support hot reloading for this. Hence, the user will need to run cmake and recompile again for the script to be recognized.

6.3 Tags

The bottom section displays the tags tagged to this entity. It first displays all active tags on this entity, then a dropdown for the user to add or remove tags on this entity.

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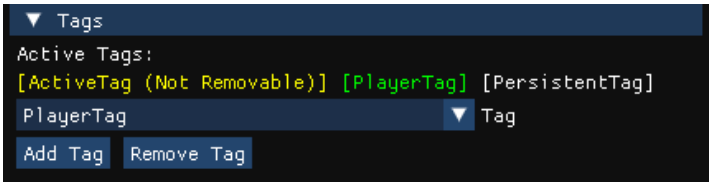


Figure 15: Selecting an attached tag

7.0 Asset Editor (Top Right)

The Asset Editor panel on the top right of the editor allows the user to see what asset they have selected from the Asset Browser, including the file path, and a larger preview of the image.

Users also have the option to reimport or replace the selected asset, as shown below.



Figure 16: Asset Editor

Reimporting a file will reload the file from the disk.

Replacing a file will allow users to drag and drop a file of the same extension into the drop zone. This preserves the old file name but updates the file with the contents of the new file. Currently, data files (json/txt), font files (ttf), and texture files (pngs/jpgs) are supported for replacement.

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8.0 Layer Editor/Viewer (Top Right)

The Layer Editor/Viewer is split into three sections:

- Scene Entity Layer Tracker
- Collision Matrix
- Entity Layer Configuration

8.1 Scene Entity Layer Tracker

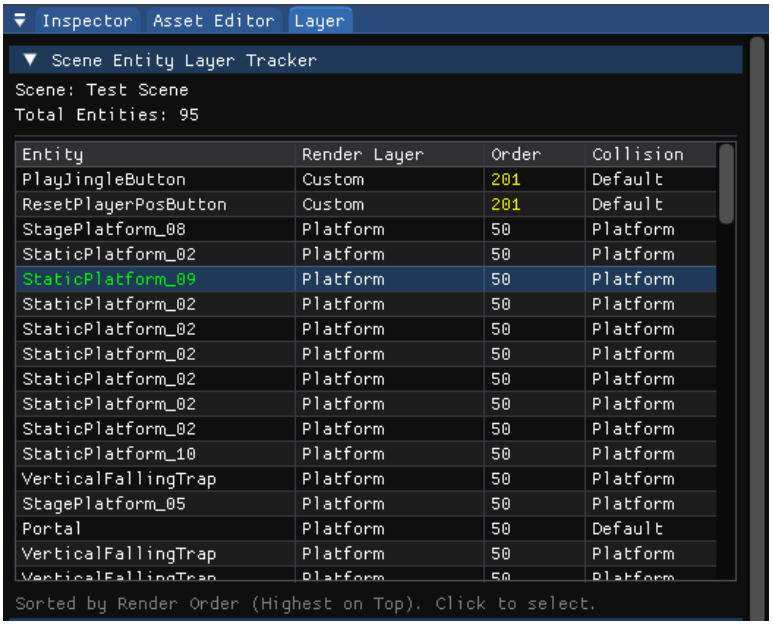


Figure 17: Scene Entity Layer Tracker

8.2 Collision Matrix

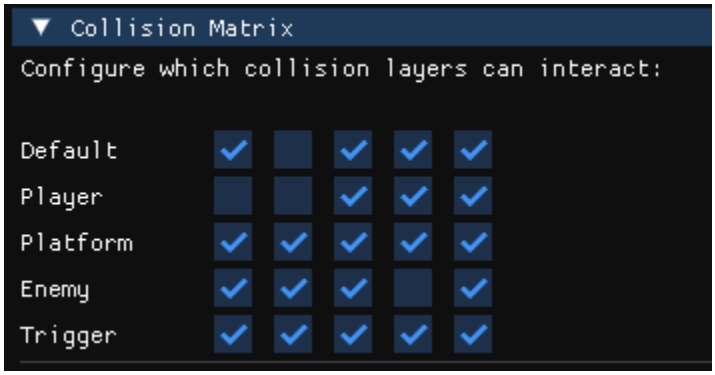


Figure 18: Collision Matrix

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8.3 Entity Layer Configuration

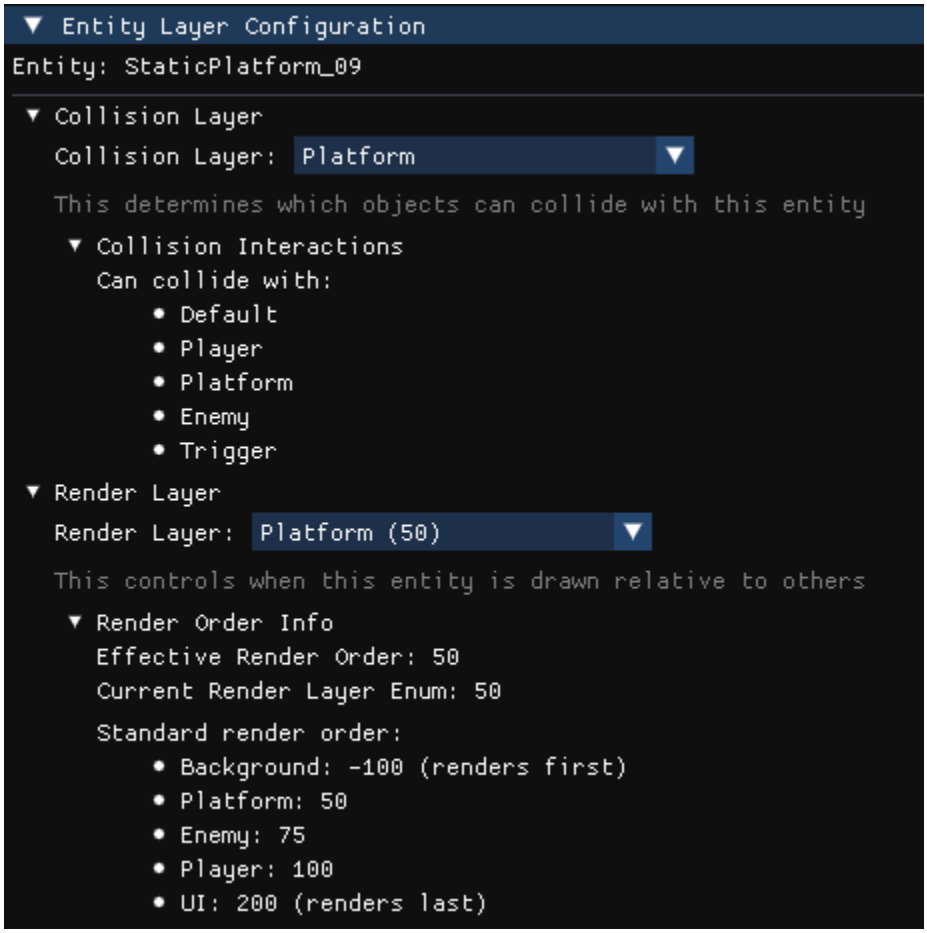


Figure 19: Entity Layer Configuration

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9.0 Performance Profiler (Bottom Right)

The performance profiler is a critical tool for optimization and identifying bottlenecks.



Figure 20: Performance Profiler

As seen above in Figure 20, the update intervals can be adjusted using the slider for debugging purposes. Below that, these are displayed:

- **Total Frame Time**
    - This shows the total frame time and in brackets, the frames per second.
- **System Breakdowns**
    - Here, it lists all systems and shows the processing time that each system is consuming. This is a debugging feature that allows developers to pinpoint which system is consuming the most processing time, guiding optimization efforts.

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